

Kraeusening

From German brewing and more

Besides sugar and dried malt extract (DME), beer can also be carbonated with unfermented wort (a.k.a. *Speise*) or actively fermenting beer (*Kräusen* or *Kräusen* beer, sounds like croysen). This procedure is generally used by breweries which bottle condition their beers. The reason for this is simple: freshly fermenting beer is readily available in a brewery. It also has several advantages over priming with corn sugar/DME:

- The flavor of the beer will not be changed, since a part of the original wort (same IBUs, same malt profile, etc.) is added
- The apparent OE (original extract) or starting gravity will not be changed. If you use a highly concentrated sugar solution your FG and ABV will be as if the beer had a .5 – 0.75 Plato (2-3 gravity points) higher starting extract.
- In the case of *Kräusening* fresh and healthy yeast is added which does a better job scrubbing oxygen and off-flavors from the beer. In this case you can also use your bottles as a yeast bank since it is not the least flocculant yeast that is at the bottom of those bottles
- Carbonation time will be more predictable, because fresh yeast is added. Since carbonation takes only a few days to a week, as opposed to many weeks, the beer can be aged longer in bulk while it takes the same amount of time from kettle to glass.
- You will have more beer to bottle and can get 5 gal out of using a 5 gal carboy for fermentation.

The main disadvantage is added complexity.

Contents

- 1 Preparation
- 2 Calculating the volume
- 3 Bottling day
- 4 Conditioning
- 5 Kegging
- 6 Final Comments

Preparation

If you want to prime with *Speise* or *Kräusen*, you are best off keeping some of the original wort from that batch. Some brewers just keep it in a sanitized jar in the fridge, but I like to store mine in the freezer. This way I don't have to worry about contamination during the next few weeks until I bottle. I also don't worry about sanitation when I collect the wort. As shown in the Whirlpooling article, I get it by filtering the hot-break/hop sludge, that is left in the kettle, through a paper towel.



When I need it, I will take the wort from the freezer defrost and then boil it for 10 min in a 2000ml Erlenmeyer flask. This will sanitize it. If you need to add more bitterness to the beer you can also add hops to this boil and boil longer. Chill the wort overnight or in an ice bath.

If you want to bottle with *Kräusen*, you need to inoculate the wort with yeast. You may add dry or liquid yeast. I simply take a sanitized racking cane to pull some yeast samples from the fermenter. To do this, close one end of the cane with your thumb and push it all the way to the bottom of the primary or secondary. Open it to suck up some of the yeast sediment. Moving it around while doing this also helps. Close it up again and move the cane to your starter vessel and dump the yeast in there. Repeat this a few times until you have enough yeast.

At this point you may also use a different yeast than the one used for fermenting the beer. This can provide a few advantages. Lager yeast, for example, may be used during the colder months to allow conditioning of an ale at basement temperatures. Better flocculating yeast can be used to bottle a beer which has been fermented with poorly flocculating

yeast. In this case it is beneficial to drop out the initial yeast with a fining agent like gelatin. My Kaiser Alt is bottled this way.

Now wait until it starts fermenting.

Calculating the volume

The problem with calculating the necessary volume of *Kräusen* arises from the fact that it's sugar content keeps changing while it is fermenting. As a result you'll have to make a guess at the needed volume when you prepare it and calculate the actual volume on bottling day.

Before the necessary *Kräusen* volume can be calculated the amount of missing carbonation needs to be calculated. There are two different units of measurement which are commonly used for carbonation. g/l CO₂ expresses the weight of the dissolved CO₂ in one liter of beer whereas volumes of CO₂ is a bit more complicated to explain. Volumes of CO₂ expresses how much larger than the beer volume the CO₂ volume would be if it was not dissolved in the beer but a gas at atmospheric pressure. Here is an example. 1 pint of beer with a carbonation of 2 volumes of CO₂ means that the CO₂ in that pint of beer would occupy 2 pints if it was not dissolved in the beer but captured in a non pressurized bag. The conversion between the two is simple: One volume of CO₂ equals 2 g/l CO₂.

The amount of CO₂ that needs to be added to the beer depends on two factors: the current CO₂ content of the beer and the desired carbonation. The CO₂ content of "still", i.e. unpressurized beer with a head space full of CO₂, depends largely on its temperature. It can be found in Carbonation Tables in the column which corresponds to 0 pressure (either kPa or psi).

The desired carbonation depends on the style of beer and brewer's preference. The aforementioned Carbonation Tables list carbonation guidelines for various styles. The missing carbonation, which needs to be supplied by the sugar in the *Kräusen*, is the difference between the current CO₂ content and the desired CO₂ content.

With the assumption that 1 g of sugar is fermented into equal parts of CO₂ and ethanol, which is not exactly true but good enough for our calculations, the following formulas calculate the necessary *Kräusen* volume. This calculation makes the assumption that the *Kräusen* has been prepared with the same wort as the batch of beer to be primed:

For metric units:

$$V_k = \frac{0.24 \cdot V_{pb} \cdot c_k}{(AE_k - AE_b)}$$

where

- **V_k**: *Kräusen* volume in liter
- **V_{pb}**: volume of the primed beer in liters. The volume of the primed beer includes the *Kräusen* volume. See text for further explanation
- **c_k**: the carbonation that needs to be provided by the *Kräusen* in g/l CO₂
- **AE_k**: the current apparent extract of the *Kräusen* in Plato. This is taken from a hydrometer reading
- **AE_b**: the current apparent extract of the beer. This is taken from a hydrometer reading and assumes that the beer has completed fermentation. I.e. this is the final extract of both the beer and the *Kräusen*.

for U.S. units:

$$V_k = \frac{1.95 \cdot V_{pb} \cdot c_k}{(GU_k - GU_b)}$$

where

- **V_k**: *Kräusen* volume in quarts

- V_{pb} : volume of the primed beer in quarts. The volume of the primed beer includes the *Kräusen* volume. See text for further explanation
- c_k : the carbonation that needs to be provided by the *Kräusen* in volumes CO₂
- GU_k : the current gravity units (points) of the *Kräusen*. This is taken from a hydrometer reading
- GU_b : the current gravity units of the beer. This is taken from a hydrometer reading and assumes that the beer has completed fermentation. I.e. this is the final gravity of both the beer and the *Kräusen*.

One of the inputs to these formulas is the volume of the primed beer (V_{pb}) which includes the *Kräusen* volume that is the result of the calculation. This is done to keep the formula simple and the best way to handle this, is to make an initial guess at the *Kräusen* volume. 5% of the beer volume works well. If the calculated *Kräusen* volume is then much different than the initial guess the calculation can be repeated with the better *Kräusen* volume estimate. After that it should be accurate enough.

If you want to calculate the amount of *Speise* necessary simply use the original extract or starting gravity as AE_k or GU_k , respectively.

On occasion it can happen that the calculated volume of *Kräusen* is more than what you have available. Especially, since you need to leave some of the *Kräusen* behind in order to avoid adding the yeast sediment to the beer. In these cases simply calculate the carbonation you can get from the *Kräusen* volume available to you and add sugar for any missing carbonation.

These are the formulas that allow you to calculate the carbonation you are getting from the available *Kräusen* volume:

metric units:

$$c_k = \frac{4.17 \cdot V_k \cdot (AE_k - AE_b)}{V_{pb}}$$

US units:

$$c_k = \frac{0.51 \cdot V_k \cdot (GU_k - GU_b)}{V_{pb}}$$

Once you know the carbonation you get from the *Kräusen* you can calculate the remaining carbonation which needs to come from additional sugar:

$$c_s = c_{needed} - c_k$$

and calculate the weight of sugar with:

metric units:

$$m_s = \frac{2 \cdot c_s \cdot V_{pb}}{A}$$

where:

- m_s : sugar weight in g
- c_s : carbonation that needs to be created from the sugar in g/l CO₂
- A : the sugar's fermentability factor (1.0 for table sugar, 0.92 for corn sugar [Palmer], 0.65 for dried malt extract)

US units:

$$m_s = \frac{0.14 \cdot c_s \cdot V_{pb}}{A}$$

where:

- m_s : sugar weight in oz
- c_s : carbonation that needs to be created from the sugar in volumes CO₂
- A: the sugar's fermentability factor (1.0 for table sugar, 0.92 for corn sugar [Palmer], 0.65 for dried malt extract)

To simplify the calculations, support priming with *Kräusen* that has not been prepared from the same wort as the beer and allow for the consideration of residual fermentable extract in the beer to be primed I created a spreadsheet: carbonation calculator US units (http://braukaiser.com/documents/Kaiser_carbonation_calculator_US.xls)/ carbonation calculator metric units (http://braukaiser.com/documents/Kaiser_carbonation_calculator_metric.xls). You can use this instead of the aforementioned formulas to determine how much *Kräusen* and sugar is needed to achieve the targeted carbonation.

Bottling day

On bottling day you will need to do the following:

- Take a gravity reading from the beer that you are going to bottle
- With a sanitized spoon skim the *Kräusen* off the *Kräusen* beer. You don't want that brown gunk in your beer.
- Take a gravity reading of the *Kräusen* beer
- Fill-out the spread sheet and determine the amount of *Kräusen* beer needed. Note that you will not be able to use all the *Kräusen* beer, since you want to hold back the sediment. If you need more than you have calculate the necessary sugar addition.
- Boil sugar if needed
- With one pour (no tilting back) pour the *Kräusen* beer into a sanitized measuring cup until the desired volume is reached. Make sure that you get as little as sediment as possible. The yeast in suspension will be plenty for carbonation
- Pour or gently place the *Kräusen* beer into the bottling bucket
- Rack the beer on top of it. Also make sure that it mixes well with the *Kräusen*.
- Once you have enough beer in the bottling bucket, add the sugar solution if necessary. Adding it too early could scald the yeast
- If necessary mix the contents of the bottling bucket. I gently stir with the sanitized racking cane or turkey baster which I use for sampling.
- Bottle as usual.

Conditioning

The bottles should carbonate within about a week at the fermentation temperature appropriate for the yeast (10-14 C / 50-58 F for lager and 18-21 C / 64-70F for ales). This quick conditioning time is one of the benefits of *Kräusen*ing. The beer may still benefit from additional aging though.

Kegging

I may use the same method for carbonating my lagers in a corny keg before I start lagering. But in this case I just make sure that enough *Kräusen* is added to get above the desired CO₂ level. I then monitor the pressure build-up and blow-off any excess CO₂. This has the advantage of adding fresh and healthy yeast which may attenuate the beer better before the cold conditioning is done. After cold conditioning, I rack the beer off the sediment into a serving keg.

Final Comments

Is all this extra work worth the effort? Every brewer has to decide this for himself/herself. Though it seems more complicated, the additional steps don't add much overhead to bottling for me since most of the time is spent filling bottles anyway. I like that the beer will be ready within about a week after bottling. Using the spreadsheet and measuring the actual attenuation of the *Kräusen* on bottling day removed the worry of having to wait until the *Kräusen* beer reached just the right attenuation.

Retrieved from "<http://braukaiser.com/wiki/index.php?title=Kraeusening&oldid=3932>"

- This page was last modified on 2 January 2010, at 16:12.
- This page has been accessed 43,966 times.
- Content is available under Attribution-NonCommercial 3.0 Unported unless otherwise noted.